Final Report: Bitcoin Proof Explorer

Student Information

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Project Title

Bitcoin Proof Explorer

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1. Introduction

This project implements a modular dashboard for exploring the Bitcoin blockchain, focusing on the Proof of Work (PoW) mechanism, block structure, difficulty adjustment, and anomaly detection using AI. The dashboard is built with Streamlit and integrates live data from the Blockstream API. The goal is to provide an educational, interactive, and visually clear tool for understanding how Bitcoin's security and consensus work in practice.

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2. Project Structure

The project is organized as follows:

- **api/**: Blockchain API client and data fetchers.
- **modules/**: Each module (M1–M4) is implemented as a separate Python file.
- **app.py**: Main Streamlit dashboard, navigation, and UI logic.
- **report/**: Contains this final report.
- **requirements.txt**: Python dependencies.

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3. Module Overview

M1: Proof of Work Monitor

- **Description**: Displays live Bitcoin network data, including latest block height, hash, difficulty, estimated hash rate, and block intervals.
- **Features**: Real-time updates, interval charts, and explanatory notes.
- **Purpose**: Helps users understand how PoW secures the network and how block intervals fluctuate.

M2: Block Header Analyzer

- **Description**: Parses and explains the 80-byte Bitcoin block header, showing all fields (version, previous hash, Merkle root, timestamp, bits, nonce).
- **Features**: Local verification of PoW using hashlib, field-by-field explanations.
- **Purpose**: Demystifies the structure of a block and the cryptographic link between blocks.

M3: Difficulty History

- **Description**: Loads and visualizes recent difficulty adjustment periods (2016 blocks per period).
- **Features**: Plots actual vs. target block times, shows how difficulty adapts to maintain a 10-minute average.
- **Purpose**: Explains the self-regulating nature of Bitcoin's mining difficulty.

M4: AI Component – Anomaly Detection

- **Description**: Implements statistical and machine learning methods to detect unusual block intervals.
- **Features**:
  - **Baseline**: Z-score calculation for block interval anomalies.
  - **AI Model**: Isolation Forest trained on live block timing features.
  - **Extra Feature – Anomaly Traffic Light**: A "traffic light" (semaforo) system visually summarizes the anomaly status of recent blocks:
    - **Green**: No significant anomalies detected.
    - **Yellow**: Watch – some intervals are statistically unusual.
    - **Red**: Tension – strong anomaly signals detected by both baseline and AI.
  - **Candidate Table**: Ranks blocks by anomaly score for inspection.
- **Purpose**: Demonstrates how unsupervised AI can complement traditional statistics in blockchain monitoring, and provides an intuitive, at-a-glance summary for users.

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4. Extra Feature: Anomaly Traffic Light (Semaforo)

A key enhancement in this project is the "anomaly traffic light" system. This feature aggregates the results of both the statistical baseline and the AI model to provide a simple, color-coded indicator of network health:

- **How it works**:
  For each new block, the system computes both the z-score and the Isolation Forest anomaly score. If both methods agree on a significant anomaly, the light turns red. If only one method signals an anomaly, it turns yellow. If neither detects an issue, it stays green.
- **Why it matters**:
  This approach makes complex anomaly detection results accessible to non-experts, allowing quick assessment of whether the network is behaving as expected or if further investigation is needed.

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5. Results and Discussion

- All modules are fully functional and integrated into a single dashboard.
- The dashboard updates with live data and provides clear explanations for each concept.
- The anomaly detection module successfully highlights unusual block intervals, and the traffic light system makes the results easy to interpret.
- The code is modular, well-documented, and follows the classroom template.

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6. How to Run

```
pip install -r requirements.txt
streamlit run app.py
```

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7. Conclusion

The Bitcoin Proof Explorer dashboard achieves its educational and technical goals: it guides users through the core mechanisms of Bitcoin, provides real-time insights, and demonstrates the value of combining statistical and AI methods for blockchain analysis. The anomaly traffic light system is an effective, user-friendly innovation that enhances the interpretability of the results.

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8. References

- Blockstream API documentation
- Streamlit documentation
- Scikit-learn documentation

(Add screenshots of each module and the traffic light feature for extra clarity.)